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IF5100 - Pemrograman Data Analitik

Assertion → Good defense programming

>> assert [kondisi benar], [Pesan error]

→ return [AssertionError]

Metode "cek dulu baru eksekusi".

Error in Python:

- Syntax Error
- Name Error
- Attribute Error
- TypeError
- ValueError
- IOError

```
try:
    Block of codes
except:
    Block of codes
else:
    Block of codes
finally:
    Block of codes
```

- Keywords:
  - try
  - except ~> catch errors
  - else ~> executed when no errors
  - finally ~> Execute no matter how error is.

Stop execution, signal error condition

```
>> raise Exception("descriptive string")
```

```
>> raise <exceptionName>(<arguments>)
```

OOP

- > Internal representation should be private => Tidak dapat diakses dari luar.

OOP ~> Digunakan karena mendekati objek dunia nyata
→ Dikenali melalui sifat-sifatnya (properties)

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Yang dilakukan oleh OOP:
Membuat sebuah kelas => sebagai "blueprint"
Menggunakan ~~kes~~ kelas => using instance => Object

"__init__" → to initialize some data attributes.

OOP in Python → Attributes
→ Method

↳ Bisa dilakukan override → Print → diganti pada "__str__".

Setiap kelas harus ada getter dan setter untuk setiap atribut data yang tersedia.

Override
↳ Kalau ada nama kelas yang sama maka definisi kelas yang akan digunakan adalah definisi kelas saat ini.

Class variable → Bersifat lintas objek

Aspects of Language
Semantics ← → Syntax => sentence structure
⇓
Meaning of sentence

Phython program => Sequence of definitions and commands
↳ evaluated ↳ executed

→ ~~Objek~~ Objects → Scalar => can not sub divide
→ Non-scalar => Have internal structure

<object><operator><object>
expression

typing → Strong typing
→ weak typing
→ Static typing
→ dynamic typing

input : >> input("Lore ipsum ...")

Control flow : # >> if :... else :...
>> if : ... elif : ... else : ...
>> while <condition> : ... <expression>.

for <variable> in range(<some-num>)
<expression>
<expression>.

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Exit while loop → >> break

Functions → Mechanism to abstraction and decomposition
→ documentation in docstrings
→ Characteristic : - Has name
- Has parameters
- Has a docstring (optional)
- Has a body
- Return something

- Module — contains → Executable statements
→ Function definitions

>> import <module_name>
>> from <module_name> import * → import all class's method
>> import <module_name> as [c] → aliasing

Files :
>> with open("lorem.txt", "w") as name_handle :
<expression> → name_handle. read() · write(s)
· readline() · writelines(s)
· readlines() ·
→ w, r, a

- Compound data types → List
→ Tuple
→ Dictionary

String → Sequence of char.
→ Immutable
~~test~~ len(s)
Check string length

>> Slice [Start : Stop : step]

Tuple : Ordered sequence of elements, Immutable. >> t1 = ("Apple", 2)
→ swap (x, y) = ~~(y, x)~~ (y, x)
→ return more than one value

List : Ordered sequence of elements, Mutable
→ • append => menambahkan item ke dalam list.
→ (+) => List concatenation → • remove(<items>) · ".join(L) => Join list.
→ • del (L[index])
→ • pop()
s.split(" ") => Split to list

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Sort vs sorted
→ returns sorted list only
>> sorted(L)
→ Mutate list
>> L.sort()
>> L.reverse()

Recursive process — consists of
→ Base case
→ Recursive step

Algorithm Complexity
→ Mengukur seberapa efektif dan efisien suatu algoritma.
→ Solusi : Order of Growth
→ Big O notation.

Fibonacci memoisasi vs. Fibonacci Iteratif vs Fibonacci rekursif

Common error types:
SyntaxError → Can't parse program
NameError → Variable name not found
AttributeError → Attribute reference fails
TypeError → Operand doesn't have correct type
ValueError → Operand type okay, but value is illegal
IOError → IO system reports malfunction

String.split(<delimiter>) => Split string
" ".join(List) => Join list

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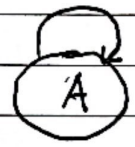
Dynamic Programing → Memoisasi

Melakukan memoisasi terhadap proses yang berulang.

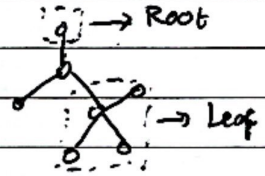
Yang berulang tidak perlu disimpan.

Graph → Undirected
→ Directed
→ Unweighted
→ Weighted



Cyclic Graph

Tree → merupakan graf khusus
→ bersifat directed



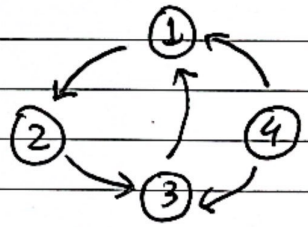
Building graphs → Nodes
→ Edges
→ ~~Stitch~~ Stitching together to make graphs

Adjacency matrix ⇝ memory intensive

	1	2	3
1	0	1	0
2	0	0	1
3	1	0	0
4	1	0	1

Source ↓

Destination →



Digraph ⇝ Directed Graph

* Graph merupakan spesialisasi dari Digraph.

BFS tidak bisa diadopsi untuk weighted karena tidak mencoba seluruh kemungkinan

Algoritma greedy → Perlu dilakukan sort !!
→ Setiap menemukan goal maka dia langsung berhenti, terlepas dari semua solusi sudah dicoba atau belum.

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